

EXPERIMENT-2

8-BIT SUBTRACTION

EXP-2

AIM: To write an assembly language program to implement 8-bit subtraction using 8085

ALGORITHM:

1. Start the program by loading the first data into the accumulator.
2. Move the data to a register.
3. Get the second data and load it into the accumulator.
4. Subtract the two register contents.
5. Check for borrow.
6. Store the difference and borrow in the memory location.
7. Halt.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 8000	Load accumulator with the first number in the address.
MOV B, A	Move the data from accumulator to B register.
LDA 8001	Load accumulator with the second number in the address.
SUB B	Subtract the data B register in accumulator.
STA 8002	Store the data (output) of the accumulator in the address.
RST 1	Halt.

INPUT:

ADDRESS	DATA
8000	4
8001	5

OUTPUT:

ADDRESS	DATA
8002	1

RESULT:

Thus the program executed successfully using 8085 processor simulator.

0101 ✓

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A01

BC0400

DE0000

HL0000

PSW0000

PC420C

SPFFFF

Int-Reg00

Flag

S0

Z0

AC0

P0

C0

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0-+00

Update Port Value

Memory

8001-+05

Update Memory

Load me at

1LDA 8000

2MOV B,A

3LDA 8001

4SUB B

5STA 8002

6RST 1

7

DataStackKeyPadMemoryI/O Ports

Start8000OK

Address (Hex)AddressData

1F4080004

1F4180015

1F4280021

1F4380030

1F4480040

1F4580050

1F4680060

1F4780070

1F4880080

1F4980090

1F4A80100

1F4B80110

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle